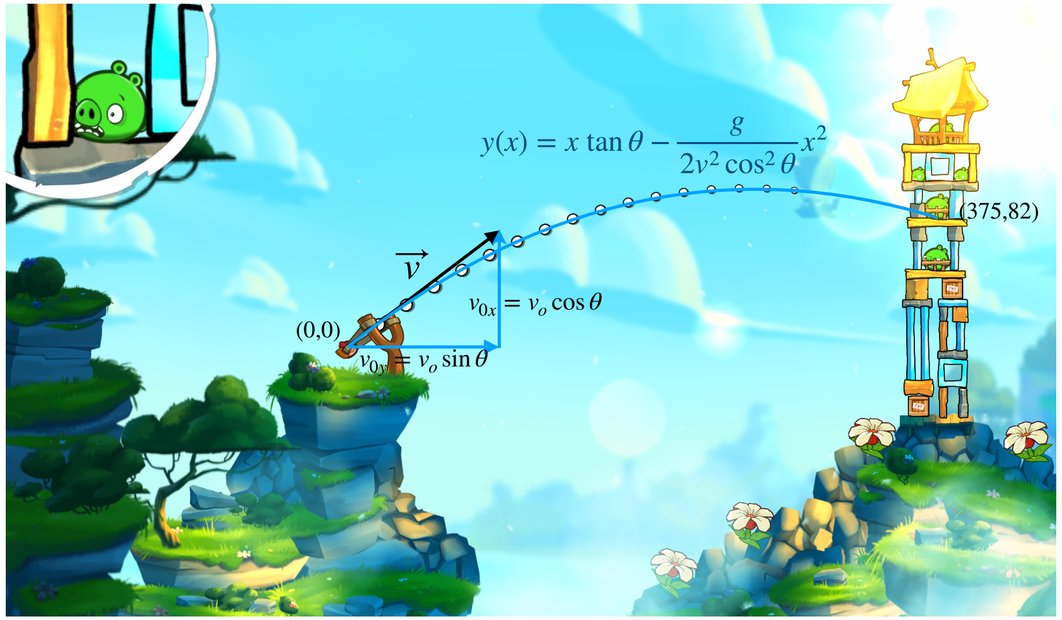
Discussion week 2



How physicists play angry birds

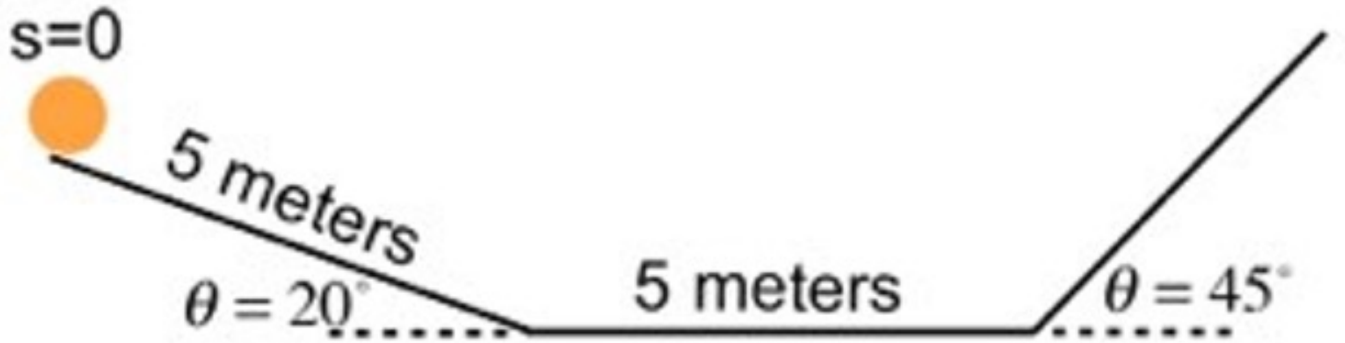
University of California, Riverside

Oct 7 , 2021

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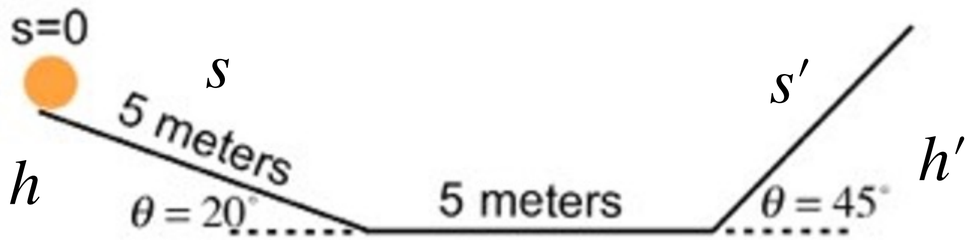
1. Ramp problem

1. A ball is placed in the following position on a ramp and released.



a. What is the acceleration of the ball down the ramp?
b. What is the velocity of the ball when it gets to the bottom of the first ramp?
c. What is the velocity of the ball when it gets to the bottom of the second ramp?
d. How far along the second ramps does the ball go before the ball stops?
e. Compare the heights of the ball at t=0 and when it stops on the second ramp (solve it!).

1.Ramp Problem



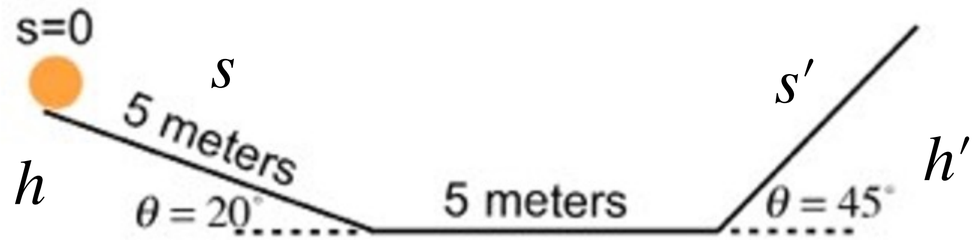
a. What is the acceleration of the ball down the ramp?

$$a = g\sin(20°) = 3.35\mathrm{m/s}^2$$

b. What is the velocity of the ball when it gets to the bottom of the first ramp?
c. What is the velocity of the ball when it gets to the bottom of the second ramp?

$$v = \sqrt{2as} = 5.79\mathrm{m/s}^2$$

1.Ramp Problem



d.How far along the second ramps does the ball go before the ball stops?

$$a' = g\sin(45^\circ)$$

$$s' = \frac{v^2}{2a'} = 2.42\text{m}$$

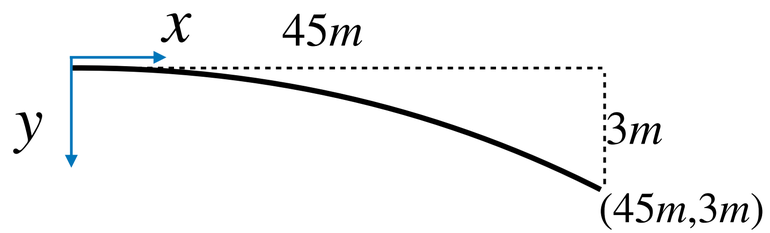
e.Compare the heights of the ball at t=0 and when it stops on the second ramp (solve it!).

$$h' = s'\sin(45^\circ) = 1.71\text{m}$$

$$h = s\sin(20^\circ) = 1.71\text{m}$$

2. Projectile Motion I

2. A rifle is aimed horizontally at a target 45.0 m away. The bullet hits the target 3.00 cm below the aim point.



$$\Delta x = v_0 t$$

$$\Delta y = \frac{1}{2} g t^2$$

a. What was the bullet's flight time?

$$t = \sqrt{\frac{2\Delta y}{g}} = 0.782\text{s}$$

b. What was the bullet's speed as it left the barrel?

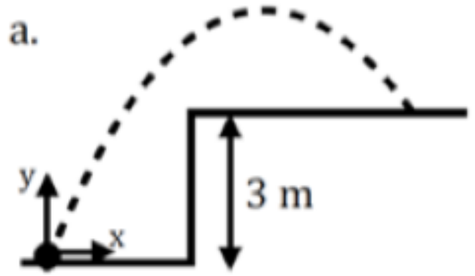
$$v_0 = \frac{\Delta x}{t} = 57.5\text{m/s}$$

3. Projectile Motion 2

1. The parts below focus on this final step of general projectile problem solving. For each part, consider a projectile fired from an initial location as shown in the accompanying figure. A coordinate system is shown in each figure, placing the origin at the projectile's initial location. Assume that you have to solve for $v_x(t)$, $v_y(t)$, $x(t)$ and $y(t)$ and would like to answer the question posed. Describe the final steps. Part a) is modeled for you.

   a. What is the horizontal range for the projectile in flight?

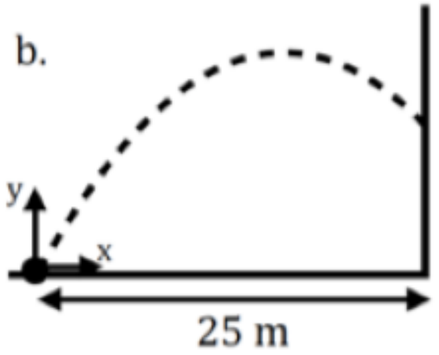
   Find the time when $y(t) = 3$ m.
   Then plug this time into $x(t)$.



   b. At what height does the projectile hit the wall?

   Find the time when $x(t) = 25$m

   Then plug this time into $y(t)$

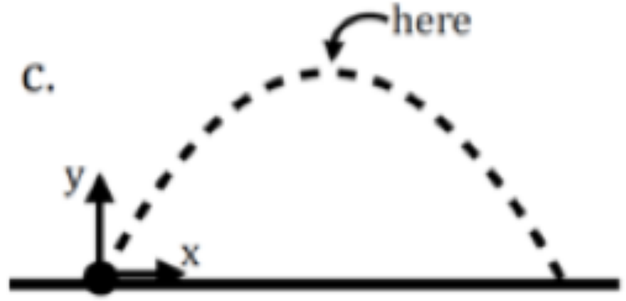


3. Projectile Motion 2

. What is the maximum height of the projectile?

Find the time that maximizes $y(t)$, i.e. $y'(t)=0$

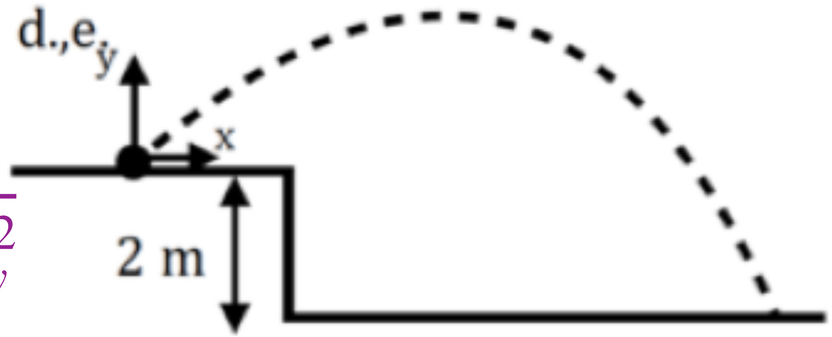
Then plug this time into $y(t)$



. What is the speed of the projectile *just before* it hits the ground.

Find the time when $y(t)=-2m$

Then plug this time into $v_x(t)$, $v_y(t)$, $v=\sqrt{v_x^2+v_y^2}$



. How long does the projectile spend in the air?

Find the time when $y(t)=-2m$